

Q-1: Given cache size = 4 blocks, Block Address = 129;
Find the cache index where this block will be mapped?

Q-2: Find the cache block address for the memory address = 321, if the block size is 10 bytes.

Q-3: Find the cache index for the memory address = 239, given that the Block size is 32 bits and the cache size is 64 bits.

Q-4: Find the cache index, Tag, and offset bits for the memory address = 297, given that the Block size is 8 bits and the cache size is 32 bits.

Q-5:

Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
010	1	011	Data
011	1	001	Data
100	0		
101	1	000	Data
110	1	010	Data
111	0		

1. 100100
2. 111010
3. 010110
4. 100100
5. 111110

- a) For the following memory access, determine if it will be a hit or miss.
- b) Calculate the hit and miss ratio.
- c) Find the AMAT if the hit time is 2s and the miss penalty is 5s.
- d) If the CPU has a 2s clock, then determine how many cycles are required per instruction.